

BRIDGEPULSE · CALCULATION

BridgePulse sizing calculations

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.6	2026-10-02	Amish Chadha	Approved decisions of 2026-10-02 carried into the design: lanyard girder clamp, pad eye and wire added to the mass (H1) and rated (H4a, H4c); BOM line 4 re-priced and installation banding added (K); R12 now \$25.00 over the value-engineering target; H3 corrected to the printed 75 Hz
0.5	2026-10-02	Amish Chadha	Lanyard anchor recorded as decided on 2026-10-02 (independent girder clamp); text only, no number changed
0.4	2026-10-01	Amish Chadha	Budget treated as a value-engineering target; R12 wording only, no number changed
0.3	2026-09-30	Amish Chadha	Design made constructable (BRP-DDR-004); mounting mass, stiffness, retention and root fillets (H1, H2, H4, H4b), FieldNode on the square post (H8), cost (K) and results (L) re-run; R12 now not met by \$2.00
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); load gating simulated (B10 to B12, C4), TMP1826 class

probes (E), 11-byte reduced summary and regional limits checked (G), cost (K), results table (L)

0.1

2026-09-25

Amish Chadha

First issue for TRL 3 (bridge dynamics, excitation and frequency precision by simulation, change detection statistics, strain, temperature, power, data and uplink, mounting stiffness and clearance, installation time, outdoor life, cost)

BridgePulse sizing calculations

On paper, BridgePulse can measure what it sets out to measure, provided it ignores the moments when people are on a light footbridge. Of its thirteen requirements, seven are met (four by calculation, three by design), two are at risk, three cannot be verified at TRL 3, and one, the cost requirement R12, is over the value-engineering target by \$25.00 since versions 0.3 and 0.6 added the parts that make the design buildable and the lanyard anchor and installation banding decided on 2026-10-02. The instrument itself is good enough: a simulation of the hourly record shows that, with a curve fit in place of simple peak picking, the scatter from sensor noise is 0.05 % at a modest 100 μg of bridge response, close to the theoretical floor. But on the 7 m example footbridge one 75 kg walker adds enough mass to lower the first mode by about 5 % while they cross, so an estimate taken over the whole record moves with who happened to walk over. Version 0.2 applies the decision recorded in BRP-DDR-002: the hub gates out each crossing, found from the strain step, and fits the rest of the record. In a time-varying simulation this cuts the scatter from 1.5 to 3.1 % to 0.015 % or less, so R2 moves from not met to at risk until recorded data confirm it. R3, the 1 % change detection, depends on how well a model can remove temperature and any remaining load effect; this note sets the target that model must meet (a daily residual of 0.75 % or less), which only field data can confirm. Version 0.2 also meets R5 on paper with a TMP1826 class probe and R7 with an 11-byte reduced summary, at a cost of \$4. The calculations also changed the mounting: the TRL 2 plate, held only at the bottom flange, would have resonated at about 75 Hz, too close to the measured band, so the plate now lies against the web between the flanges and is jacked against the top flange. Version 0.3 follows the design for construction of BRP-DDR-004: the plate is kept clear of the root fillets and stands on two clamped foot blocks, which gives 135 to 271 Hz, and the added clamp parts and glands brought the BridgePulse-specific parts to \$252.00. Version 0.6 carries in the decisions of 2026-10-02: a second, independent girder clamp with a wire lanyard as the second retention path, rated here against the falling mass with a dynamic factor, and stainless banding with one-way crimped buckles for installed units. They bring the mass on the girder to 3.58 kg and the BridgePulse-specific parts to \$275.00, \$25.00 over the \$250 value-engineering target (R12; savings in the design decisions register, BRP-DEC-001). Every number in this note is printed by `docs/04-calcs/sizing.py`; the tag in brackets, for example [A3], is the line of that script's output that carries it.

Safety: These are first-principles estimates for a paper proof of concept. They say nothing about whether any bridge is safe. BridgePulse supports inspection by qualified engineers and never replaces it. Installation is work at height, over water or near traffic, and may disturb lead paint; see BRP-PRC-001, Safety.

■ Scope and method

The note checks every requirement in BRP-REQ-001 v0.8 against the design in BRP-PRC-001 v0.8 and the parametric model `cad/src/model.py`. The script imports the model's `PARAMS`, `derived()` and part solids, so the girder section, plate, hub position, clearances and part volumes used here are the ones in the STEP files and in drawing BRP-DWG-001. It also reads `bom/bom.csv` and `budget_usd` in `project.yaml`. Run it from the repo root with `python docs/04-calcs/sizing.py`; it takes about 20 s, most of it the simulations in section B, which use fixed random seeds so its figures repeat exactly.

The design case is the example bridge of BRP-PRC-001: a 7 m steel footbridge, 6.4 m between bearings, on two IPE 360 class girders at 1.2 m centers with a 50 mm timber deck, monitored at midspan, with a FieldNode core on the midspan handrail post reporting hourly over LoRaWAN.

■ Assumptions

Table 1. Main assumptions.

Area	Assumption	Basis
Bridge	Simply supported, pinned bearings, no composite action from the deck; girder section from the model's rectangles (no root fillets); steel 210 GPa, 7,850 kg/m ³ ; timber deck 500 kg/m ³ ; rails and posts 20 kg/m	Screening model; the catalog IPE 360 is slightly stiffer [A1]
Temperature	Steel modulus falls by 0.03 % per kelvin	Typical for structural steel near room temperature; assumed
Damping	1 % of critical in the first vertical mode	Assumed; light steel footbridges are often lower, which would sharpen the peak
Footfall	75 kg walker at 2 Hz and 1.3 m/s; an effective heel-strike impulse of 2 N·s on a mode above 10 Hz	Order-of-magnitude assumption, to be measured
Accelerometer	ADXL355 class, 22.5 µg/VHz, white; 250 Hz output data rate with the filter corner at 62.5 Hz	Maker's product page (BRP-PRB-001)
Spectra	Hann window, 16,384-point segments, 50 % overlap over a 600 s record	As BRP-PRC-001
Strain	Gauge factor 2.0; 350 Ω gauges; 3.3 V excitation; ADC noise 0.15 µV RMS at 20 SPS and gain 128; gauge-to-dummy thermal mismatch 1 µε/K	ADS1220 class; the noise figure is assumed because the maker's product page does not state one
Power	5 V switched rail from the FieldNode port, 85 % buck in the hub, 90 % FieldNode rail converters; the hub is off between records	FieldNode D5 and FND-CAL-001
Radio	13 bytes of LoRaWAN overhead; 125 kHz, coding rate 4/5, 8-symbol preamble, explicit header; regional payload limits of the LoRaWAN Regional Parameters (EU868, US915, AS923)	EU868 (51 bytes at DR0 to DR2) and US915 (11 bytes at DR0, 53 at DR1) checked against The Things Network's EU868 and US915 pages; the AS923 dwell-time limits are assumed
Load gating	Crossing found from the strain step and gated out with 0.25 s either side, edges tapered over 0.1 s; ambient (wind) response 10 µg RMS with nobody on the span; single mode, walker as a moving mass and heel-strike train	Assumed; the ambient level stands for a quiet bridge and the footfall model is idealized
Lanyard	3 mm 7 × 7 stainless wire rope: metallic area 53 % of the circle, effective modulus 100 GPa, minimum breaking load 4.8 kN; fitted with 25 mm of slack at most; girder clamp maker's design factor 4:1 between working load limit and ultimate load	Typical catalogue values for the class of part, assumed; to be confirmed against the bought parts
Wind	35 m/s gust, air 1.225 kg/m ³ , drag 1.3 on the FieldNode enclosure; 52.2 N on its panel	FND-CAL-001

■ A. Example bridge dynamics (R2, R3)

- **Section and mass.** One girder from the model's rectangles has $I = 15,524 \text{ cm}^4$ and $W = 862 \text{ cm}^3$, about 5 % below the catalog IPE 360 [A1]. The span carries 167 kg/m, and EI for the two girders is $6.52 \times 10^7 \text{ N}\cdot\text{m}^2$ [A2].
- **First mode.** $f_1 = (\pi / 2L^2) \sqrt{EI / m} = 23.9 \text{ Hz}$; the second vertical mode is 96 Hz, outside the band, so only the first vertical mode (and possibly a torsional mode) is tracked on this span. The modal mass is 535 kg [A3]. The TRL 2 estimate of 24 Hz stands.
- **What a 1 % shift means.** Frequency goes with the square root of stiffness, so a lasting 1 % drop in frequency is a 2 % loss of bending stiffness [A4].
- **Temperature.** The change of the steel modulus alone moves the frequency by 0.015 % per kelvin, 1.05 % over the -20 to +50 °C range of the requirements [A5]. That is as large as the target, so compensation is needed even before bearings, deck and surfacing, whose effects cannot be computed on paper.
- **People on the deck.** The example bridge is light, so the people on it change its frequency. One walker at midspan lowers f_1 by 6.3 %, and by 3.3 % averaged over a crossing; a crowd of one person per square meter lowers it by 22.7 % [A6]. On a road bridge with 20 t of modal mass one walker moves it by only 0.19 % [A7]. Mass loading is therefore a footbridge problem first.

■ B. Excitation, noise and frequency precision (R1, R2)

- **Footfall.** Each heel strike rings the 23.9 Hz mode up to about 57 mg, decaying with a 0.66 s time constant; while someone walks near midspan the response is about 29 mg RMS. A crossing takes 4.9 s [B1].
- **Resolution and noise.** The 600 s record gives 17 averaged segments with a bin width of 0.0153 Hz [B2]. Sensor noise is 2.8 μg in one bin and 177 μg RMS over the whole band [B3]. The mode's half-power bandwidth is 0.48 Hz, 31 bins wide, so the bin width is not what limits precision; a peak 10 dB above the noise needs about 62 μg RMS of modal response [B4].
- **Simulation.** The script simulates 60 hourly records at each of five levels of random modal response, adds the accelerometer's noise, and estimates the frequency two ways: by picking the highest bin and interpolating, as the TRL 2 concept proposed, and by fitting a single-degree-of-freedom spectrum to the 3 Hz around the peak.

Table 2. Scatter of the hourly frequency estimate from sensor noise alone, 1σ [B5].

Modal response (RMS)	Peak found	Peak picking	SDOF curve fit
10 μg	20 % of records	0.807 %	0.488 %
30 μg	100 %	0.359 %	0.103 %
100 μg	100 %	0.265 %	0.053 %
300 μg	100 %	0.264 %	0.040 %
1,000 μg	100 %	0.220 %	0.032 %

- **Change of method.** Peak picking never reaches the 0.2 % of R2, however strong the signal, because the averaged spectrum is still noisy across the 31 bins of the peak. The curve fit reaches 0.2 % from 30 μg RMS [B6] and approaches the theoretical floor for a randomly excited mode, $\sqrt{\zeta / 2\pi f T} = 0.033 \text{ %}$ [B6a]. The reduce step in BRP-PRC-001 now uses the curve fit.
- **Mass loading.** Because each crossing both excites the mode and loads it, an estimate over the whole record is weighted toward the moments when someone is near midspan: an energy-weighted crossing lowers it by 5.3 %, and

and the spread of walker masses (50 to 100 kg) adds 1.01 % of scatter per crossing [B7]. On that simple count the scatter falls with traffic, to 0.45 % at 5 crossings during the 10 minute record and 0.23 % at 20 [B8]; v0.1 wrote these as crossings an hour, but the count is of crossings inside the record. Ungated, R2 would need about 26 single crossings in every record, and fails outright when two people cross together, which biases that record by 10.5 % [B9].

- **Load gating (BRP-DDR-002).** The hub now drops the part of each record when someone is on the span, found from the strain step (one walker gives $3.2 \mu\epsilon$ against $0.09 \mu\epsilon$ resolution, section D), with 0.25 s either side, and fits the rest. What is left is the free decay after each crossing, at the unloaded frequency, plus ambient response. The script simulates this with a time-varying single-mode model, stepping the walker's mass and heel strikes across the span sample by sample.

Table 2a. Hourly estimate with and without load gating, 30 simulated records per case [B10].

Crossings in the record	Ungated bias	Ungated scatter	Gated bias	Gated scatter (1σ)	Records accepted	Record gated out
1	-4.73 %	3.06 %	-0.030 %	0.015 %	80 %	0.9 %
5	-5.16 %	2.50 %	-0.037 %	0.010 %	100 %	4.4 %
20	-6.26 %	1.48 %	-0.039 %	0.007 %	100 %	16.6 %

- **What the simulation shows.** The time-varying simulation gives a larger ungated scatter than the simple count of B8, because the frequency also moves with where the walker is. Gated, the scatter is 0.015 % or less, well inside the 0.2 % of R2, and the small bias moves by only 0.010 % between 1 and 20 crossings [B12]. The hub accepts a fit only if the peak stands 10 dB above the fitted noise and the fitted damping is 0.2 to 5 %; with one crossing and a quiet $10 \mu\text{g}$ ambient response, 20 % of records fail that test and are reported as "no clear peak", not as a shift [B11]. An hour with nobody crossing on a quiet day gives no estimate at all.
- **R2 is at risk**, no longer not met. The target is unchanged, as BRP-DDR-002 requires, and the method must be checked on recorded data (TRL 4, on hold). The simulation is optimistic: one mode, an idealized walker, a perfect gate and no wind during crossings. The load regression in TwinKit (BRP-DDR-002) remains as a second line for any load effect the gate misses. R1 is met by design on the datasheet.

■ C. Change detection statistics (R3)

- **Test.** One false flag per year from a daily test needs a one-sided threshold of $z = 2.78$, and 90 % detection adds $z = 1.28$ [C1].
- **What the temperature and load model must achieve.** Comparing a moving mean of daily frequencies with the baseline, the daily residual left after compensation must be no larger than the values in Table 3 [C2].

Table 3. Largest daily residual (1σ) that still detects a 1 % shift [C2].

Window after the shift	4-week baseline	1-year baseline
7 days	0.58 %	0.65 %
14 days (R3)	0.75 %	0.90 %

- **Budget.** A 0.5 K temperature reading error costs only 0.0075 % [C3]. Random walker scatter averages down to 0.09 % over a day at 5 crossings a record, but a systematic change in traffic does not: a 10 % change in the share of hours with two people on the span shifts the daily mean by 0.53 % [C3], most of the 0.75 % allowance, if records are not gated. With load gating that traffic term falls to the 0.010 % bias span of B12 [C4], leaving the allowance to temperature and the rest of the model. The statistics also assume independent days, which is optimistic.
- **R3 cannot be verified at TRL 3.** It is achievable if temperature and load together can be modeled to a daily residual of 0.75 % or less. Whether they can is a question for field data or for public benchmark data such as the Z24 record; no such data were analyzed in this session.

■ D. Strain (R4)

- **Sensitivity.** A half-bridge with one active and one dummy gauge gives 1.65 μV per $\mu\epsilon$ at 3.3 V [D1].
- **Resolution and range.** At the assumed ADC noise, one sample resolves 0.09 $\mu\epsilon$ RMS, 0.60 $\mu\epsilon$ peak to peak, well inside the 2 $\mu\epsilon$ target; $\pm 1,000 \mu\epsilon$ is $\pm 1.65 \text{ mV}$, 6 % of the $\pm 25.8 \text{ mV}$ full scale at gain 128 [D2], so gain can come down if a bridge strains more.
- **What there is to see.** A 5 kN/m² crowd gives 19.2 kN·m per girder, 22.3 MPa, 106 $\mu\epsilon$; one walker at midspan gives 3.2 $\mu\epsilon$ [D3]. The TRL 2 figures (about 100 $\mu\epsilon$ and 3 $\mu\epsilon$) stand. The strain channel is therefore a good load indicator for the gating proposed in section B, but only just for one person.
- **Aliasing.** Dynamic strain in the 23.9 Hz mode is 1.08 $\mu\epsilon$ at the heel-strike peak; sampled at 20 SPS it would alias to 3.9 Hz but stays below the resolution [D4].
- **Drift.** A 1 $\mu\epsilon/\text{K}$ mismatch between gauge and dummy with 2 K between flange and coupon gives 2 $\mu\epsilon$ of apparent strain [D5], and each gauge dissipates 7.8 mW, so a 20 s warm-up now precedes each record [D6]. Long-term drift of a coated foil gauge outdoors cannot be estimated on paper. **R4 is not verifiable at TRL 3** for drift; resolution and range are met on paper.

■ E. Temperature (R5)

- Under BRP-DDR-002 the probes are now TMP1826 class 1-Wire sensors potted in a stainless sheath. The maker specifies the WSON package at $\pm 0.3 \text{ }^\circ\text{C}$ from -40 to $+105 \text{ }^\circ\text{C}$ and $\pm 0.2 \text{ }^\circ\text{C}$ from $+10$ to $+45 \text{ }^\circ\text{C}$ (Texas Instruments) [E1]. The DS18B20 class probe of v0.1 was specified at $\pm 0.5 \text{ }^\circ\text{C}$ only from $-10 \text{ }^\circ\text{C}$ upward. The effect of a 0.5 K error on the frequency compensation is negligible in any case [E2].
- CalRig's chamber does not reach below about 9 $^\circ\text{C}$ (CLR-CAL-001), so each probe gets an ice-point check at 0 $^\circ\text{C}$ and a comparison against a reference in CalRig [E3]. **R5 is met on paper** on the datasheet; the checks are TRL 4 work.

■ F. Power (R6)

Table 4. Loads while recording [F1].

Load	mW at 3.3 V
Microcontroller at reduced clock	33.0
Accelerometer	0.7
24-bit ADC	1.4
Gauge excitation, two 700 Ω half-bridges	31.1
Bridge completion resistors	1.1
microSD writes (average)	10.0
RS-485 transceiver	1.7
Total	79.0

- **Recording.** 79.0 mW at 3.3 V is 92.9 mW at the FieldNode port through the 85 % buck [F2]. The TRL 2 figure of 84 mW did not count the converter or the RS-485 link.
- **Average.** Each hour the hub records for 620 s (with the warm-up) and processes for 60 s; then FieldNode switches the rail off, so the hub has no sleep current. The average is **16.9 mW** at the port, 18.8 mW from the FieldNode cell, 0.41 Wh a day [F3]. The TRL 2 figure of about 15 mW stands within its precision. Continuous recording would need 93 mW, and the hourly windows use 17 % of FieldNode's 100 mW sensor allowance [F4]. **R6 is met on paper.**
- **Processing.** The FFTs take about 29 Mflop, about 29 s at an assumed 1 Mflop/s in software floating point, plus about 6 s to read the record back from the card; working one axis at a time from the card needs 224 kB of the RP2040's 264 kB of RAM [F5]. Computing all three axes live would not fit, which is why the hub post-processes from the card.

■ G. Data, storage and uplink (R7)

- **Raw data.** 1.90 MB per hour, 46 MB per day, 1.39 GB per month; a 32 GB card holds 22 months [G1], writing 16.7 GB (0.52 full-card writes) a year [G2]. An industrial card is specified for temperature, not wear.
- **Summary.** The hourly summary is 36 bytes [G3], unchanged from TRL 2.
- **Airtime.** At 49 bytes on air an uplink takes 98 ms at SF7, 329 ms at SF9, 575 ms at SF10 and 2,302 ms at SF12; 24 a day use 13.8 s at SF10 and 55.2 s at SF12, over the 30 s a day that FieldNode's R9 allows on The Things Network [G4].
- **Regional limits.** The summary fits every EU868 data rate (51 bytes at SF10 to SF12) but not US915 DR0 or AS923 DR2 with dwell time (11 bytes); both allow 53 bytes one step faster, at SF9 [G5]. The EU868 and US915 limits were checked against The Things Network's regional pages (Table 1); the AS923 limits are assumed.
- **Reduced summary (BRP-DDR-002).** At an 11-byte data rate the hub sends 11 bytes: the first frequency (2), its peak amplitude (2), the two strain means (4), the steel temperature (2) and a status byte that carries the gated seconds (1); it takes 371 ms at SF10 [G6]. The hub picks it when FieldNode reports an 11-byte limit, so the region, still open in FieldNode (FND-DDR-001, O1), no longer decides whether R7 is met. **R7 is met on paper,** with the AS923 limit assumed. On-site storage is met.

■ H. Mounting, mass and clearance (R8, R9)

- **Mass.** The hub weighs 0.99 kg, the plate 1.25 kg, the foot blocks, jaws, packers, jack block, jack screw and fixings 0.97 kg (0.47 kg of it aluminium), and the lanyard girder clamp, pad eye and wire 0.37 kg: 3.58 kg on the girder [H1]. Version 0.5 gave 3.21 kg without the lanyard parts, and version 0.2 gave 3.34 kg, counting the clamp parts as steel. The TRL 2 figure of 0.6 kg for hub and clamps was far too low. The FieldNode core weighs 2.41 kg by its own calculation, not the 1.7 kg quoted at TRL 2.
- **Stiffness of the mount.** The accelerometer must move with the girder up to 60 Hz, so the hub's own mounting resonance should be above about 120 Hz. The plate, 200 × 291 × 8 mm aluminium, lies against the web and stands on two foot blocks clamped to the bottom flange; an M12 jack screw wedges it against the top flange. With the hub 155 mm above its foot it resonates out of plane between 135 Hz (ends pinned) and 271 Hz (ends fixed) [H2] (version 0.2, with a 313 mm plate standing directly on the flange: 126 to 253 Hz). The TRL 2 arrangement, a plate clamped only to the bottom flange with the hub 190 mm up, would resonate at about 75 Hz [H3], close enough to the band to distort it. This is the main design change of this note.
- **Retention.** Each foot block is clamped to the bottom flange by a steel jaw under the flange, a packer outside the tip and an M10 bolt, and the jack screw bears on the top flange [H4]. Aluminium on painted steel needs isolating washers; the plate and foot blocks are hard anodized.
- **Lanyard, the second retention path (decided 2026-10-02, BRP-DEC-001).** A stainless pad eye screwed to the plate above the hub carries a 3 mm stainless wire lanyard, 436 mm long with a thimble eye at each end, to the eye of a bought girder clamp hooked on the bottom flange tip 159 mm along the span from the nearer foot clamp, so the two fixings share nothing [H4a]. If the mount let go, the 3.21 kg on the plate would fall until the lanyard took it up. With 25 mm of slack at most, the wire's stiffness (0.86 kN/mm) gives a peak load of 1.20 kN, a dynamic factor of 38 on the 31 N weight [H4c]. A wire lanyard is stiff, so the slack, not the mass, sets the load: the fitting rule of 25 mm matters. At a maker's usual 4:1 design factor the girder clamp needs a working load limit of 30 kg or more; the BOM specifies 100 kg or more, and the wire's assumed 4.8 kN breaking load is 4.0 times the peak. The pad eye's two M5 screws in the 8 mm plate carry the same load in shear and tension, well within an M5 stainless screw's capacity.
- **Root fillets.** A rolled IPE 360 has 18 mm root fillets between web and flanges, which the version 0.2 plate would have stood on. The plate now ends 22 mm from each flange, and each foot block has a 12 mm chamfer over the fillet; the model checks both [H4b].
- **Clearance (R9).** Below the soffit there are only the clamp jaws (10 mm, with the bolts ending inside them), the lanyard girder clamp's lower jaw (8 mm), the gauge covers (6 mm), the probe lead and clips (8 mm) and the cables that wrap the south flange (10 mm), against the 15 mm limit [H5]. The hub sits 9 mm inside the flange tip line, inside the girder outline [H6]. **R9 is met on paper.**
- **FieldNode on the post.** A 35 m/s gust puts 52.2 N on the panel and 29 N on the enclosure, a 60 N·m moment and 7.2 MPa in a 50 × 3 mm post, about a twentieth of what a 1 kN load at the rail top would cause [H7]. FieldNode's V-blocks are designed for round poles and would bear on a square post's corners. On the 50 mm square post they are left off: the back plate bears on the post's flat face, and two band clamps one size longer go round the post and through the plate's slots [H8] (BRP-DDR-004). On installed units the worm-drive bands give way to stainless banding with one-way crimped buckles (BOM line 10, decided 2026-10-02), which cannot be undone with a screwdriver, for the tamper resistance R9 asks of the post-mounted FieldNode; the bench unit keeps the worm-drive bands. Nothing in the FieldNode repo is changed here; FieldNode is asked to confirm the interface and to adopt the banding rule (BRP-DEC-001).

■ I. Installation time (R8)

Nine tasks take 320 person-minutes. The critical path is the gauge work by one fitter: access and briefing, paint removal at two sites, bonding, wiring and coating, then testing and clean up, 170 min or 2.8 h, while the second fitter fits the plate, hub, FieldNode and cables in 90 min [I1]. This assumes a lead test on an earlier visit and a fast-curing coating. **R8 cannot be verified at TRL 3** for time, which needs a trial fit; the no-drilling part is met by design.

■ J. Outdoor life (R10)

The hub (IP67) hangs in shade under the deck, and its parts are rated for the range except at the edges: the RP2040 class controller is assumed to be rated -20 to +85 °C and the industrial card -25 to +85 °C. The weak points are elsewhere: FieldNode's interior reaches 58.7 to 73.3 °C on a 45 °C day without a shield, which FND-CAL-001 reports as not met, and the life of the gauge coating is unknown [J1]. **R10 is at risk.**

■ K. Cost (R12)

The BOM has ten lines, all priced. Under R12 as redefined in BRP-DDR-001 (D1) and decided in BRP-DDR-002, the estimated cost of the BridgePulse-specific parts is \$275.00 against the \$250 value-engineering target (a hypothetical control target, not a limit), \$25.00 (10.0 %) over. The FieldNode core adds \$126.00, costed in its own repo, for a complete monitor of \$401.00, which is \$151.00 over the \$250 target [K1]. The TRL 3 changes added a separate industrial microSD card (\$20), the RS-485 and buck converter modules and the jack-screw plate; v0.2 adds \$4.00 for the TMP1826 class probes (\$6.00 each against \$4.00); v0.3 adds \$2.00 for the two probe glands (line 1) and \$2.00 for the foot blocks, jaws, packers, jack block and their screws (line 4); v0.6 adds \$19.00 to line 4 for the lanyard anchor (girder clamp \$12.00, pad eye and screws \$3.00, a made-up wire lanyard \$6.00 in place of the \$2.00 plain lanyard) and a new line 10, \$4.00, for the stainless banding and one-way buckles of installed units. **R12 is over the value-engineering target** by \$25.00 on indicative prices. Value-engineering target: USD 250.00. Estimated cost of the constructable design: USD 275.00 (USD 25.00 over the target). The savings already named in BRP-DDR-002 (a consumer high-endurance card, a cheaper accelerometer breakout that still meets R1) are listed in the value engineering section of BRP-DEC-001; they no longer close the gap on their own.

■ L. Results against every requirement

Table 5. Requirement status from this note [L].

ID	Requirement	Value	Target	Status
R12	Affordable	\$275.00 BridgePulse-specific; \$401.00 complete	\$250 value-engineering target, BridgePulse-specific parts	Over the value-engineering target by \$25.00
R2	Track natural frequencies	Bin 0.0153 Hz; sensor-noise scatter 0.053 % at 100 µg (curve fit); with load gating 0.015 % or less in simulation (1 to 20 crossings per record)	Bin 0.02 Hz; 0.2 % (1σ) over a steady day	At risk (met in simulation with gating; to be checked on recorded data)
R10	Survive outdoors	Hub in shade; FieldNode heat not met in its own repo; coating life unknown	IP67 and IP65; -20 to +50 °C; 5 years	At risk
R3	Flag structural change	Achievable if the daily residual after compensation is 0.75 % or less	1 % within 14 days; one false flag per year	Not verifiable at TRL 3
R4	Measure strain	0.09 µε RMS; ±1,000 µε uses 6 % of full scale; drift unknown	2 µε; ±1,000 µε; 5 µε per month	Not verifiable at TRL 3 (drift); resolution and range met on paper
R8	Fit without harming the structure	No drilling or welding; 2.8 h estimate	No drilling; 4 h for two people	Not verifiable at TRL 3 (time); fixing met by design
R5	Measure temperature	±0.3 °C from -40 to +105 °C on the datasheet; ice-point check	±0.5 °C over -20 to +50 °C	Met on paper (datasheet)
R7	Send and keep the data	36 bytes, or 11 bytes where the limit is 11; 22 months on 32 GB; fits every listed data rate	48 bytes within regional limits; 12 months on site	Met on paper (AS923 limit assumed)
R6	Stay within the FieldNode energy budget	16.9 mW average at the port	30 mW	Met on paper
R9	Keep clear of what passes under and over	10 mm below the soffit; hub inside the girder outline	15 mm	Met on paper
R1	Measure bridge acceleration	22.5 µg/VHz, 3 axes, 250 Hz output, 62.5 Hz filter corner	25 µg/VHz; 0.5 to 60 Hz	Met by design (datasheet)
R11	Protect privacy	No camera or microphone	Structural readings only	Met by design
R13	Present results responsibly	Wording rule in the precis; not yet implemented	No safety rating anywhere	Met by design

Counts: 0 not met, 1 over the value-engineering target (R12, \$25.00 over; v0.5: \$2.00 over), 2 at risk, 3 not verifiable at TRL 3, 4 met on paper, 3 met by design (v0.2: 0 not met, 3 at risk, 3 not verifiable, 4 met on paper, 3 met by design; v0.1: 1 not met, 4 at risk, 3 not verifiable, 2 met on paper, 3 met by design).

■ Checks against the TRL 2 figures

Table 6. TRL 2 claims (BRP-PRC-001 v0.2) against this note.

TRL 2 claim	This note	Action
First mode about 24 Hz; second about 97 Hz; EI about 6.8×10^7 N·m ² ; about 172 kg/m	23.9 Hz; 96 Hz; 6.52×10^7 N·m ² from the model's rectangles; 167 kg/m	Precis updated
Bin width about 0.015 Hz; about 17 segments; 2.8 µg in one bin; about 180 µg over the band	0.0153 Hz; 17; 2.8 µg; 177 µg	Stands
Random error of one hourly estimate about 0.01 % at 24 Hz with peak interpolation	0.22 to 0.27 % by peak picking; 0.03 to 0.05 % by curve fit	Method changed to a curve fit; precis updated
Frequency not affected by people (not considered)	One walker lowers f_1 by up to 6.3 %	R2 not met; precis updated; proposals in docs/REVIEW.md
1.65 µV per µε; crowd about 100 µε; one pedestrian about 3 µε	1.65 µV per µε; 106 µε; 3.2 µε	Stands
About 84 mW recording, about 15 mW average, 0.36 Wh per day	92.9 mW at the port; 16.9 mW; 0.41 Wh	Precis updated
1.9 MB per hour; 46 MB per day; 1.4 GB per month; about 23 months on 32 GB	1.90 MB; 46 MB; 1.39 GB; 22 months	Precis updated
About 36 bytes per uplink; regional limits not checked	36 bytes; does not fit US915 DR0 or AS923 DR2	Reduced payload proposed
Hub and clamps about 0.6 kg; FieldNode about 1.7 kg	3.58 kg on the girder with the lanyard parts (v0.5: 3.21 kg; v0.2: 3.34 kg); FieldNode 2.41 kg	Precis updated
Hub about 12 mm below the soffit	10 mm (foot clamp jaws 10 mm; lanyard clamp jaw 8 mm)	Stands
About \$342 complete; \$216 BridgePulse-specific	\$374.00 complete; \$248.00 BridgePulse-specific (v0.1: \$370.00 and \$244.00)	BOM notes and precis updated